cancellation between attraction and repulsion. Einstein realized that exactly the same reasoning holds for the entire universe. In just the same way that the attractive pull of gravity acts to slow the dancer's ascent, it also acts to slow the expansion of space. And just as the dancer can't achieve stasis—it can't hover at a fixed height—without a repulsive force to balance the usual pull of gravity, space can't be static—space can't hover at a fixed overall size—without there also being some kind of balancing repulsive force. Einstein introduced the cosmological constant because he found that with this new term included in the equations, gravity could provide just such a repulsive force.

But what physics does this mathematical term represent? What is the cosmological constant, from what is it made, and how does it manage to go against the grain of usual attractive gravity and exert a repulsive outward push? Well, the modern reading of Einstein's work—one that goes back to Lemaître—interprets the cosmological constant as an exotic form of energy that uniformly and homogeneously fills all of space. I say "exotic" because Einstein's analysis didn't specify where this energy might come from and, as we'll shortly see, the mathematical description he invoked ensured that it could not be composed of anything familiar like protons, neutrons, electrons, or photons. Physicists today invoke phrases like "the energy of space itself" or "dark energy" when discussing the meaning of Einstein's cosmological constant, because if there were a cosmological constant, space would be filled with a transparent, amorphous presence that you wouldn't be able to see directly; space filled with a cosmological constant would still look dark. (This resembles the old notion of an aether and the newer notion of a Higgs field that has acquired a nonzero value throughout space. The latter similarity is more than mere coincidence since there is an important connection between a cosmological constant and Higgs fields, which we will come to shortly.) But even without specifying the origin or identity of the cosmological constant, Einstein was able to work out its gravitational implications, and the answer he found was remarkable.

To understand it, you need to be aware of one feature of general relativity that we have yet to discuss. In Newton's approach to gravity, the strength of attraction between two objects depends solely on two things: their masses and the distance between them. The more massive the objects and the closer they are, the greater the gravitational pull they